

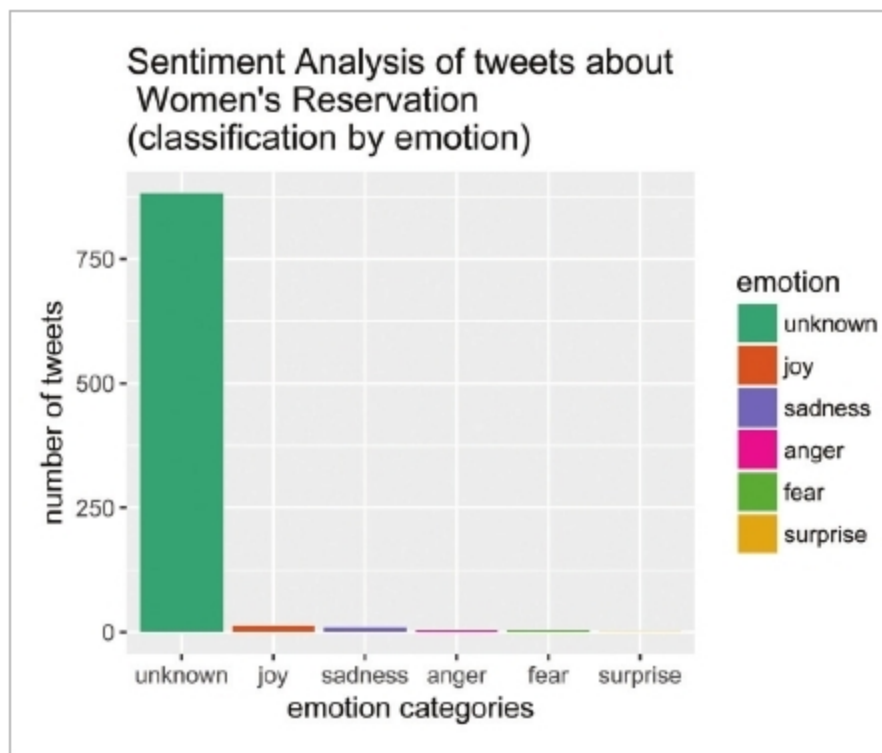


Figure 2: Histogram plot of the distribution of emotions

```
# get polarity best fit
polarity = class_pol[,4]
```

For final statistical analysis of these sentiment values, convert the corpus into the most convenient data structure of R, i.e., a data frame. This contains both the emotion and polarity values of the Twitter corpus.

```
sent_df = data.frame(text= twitter_txt, emotion=emotion,
polarity=polarity, stringsAsFactors=FALSE)

# sort data frame
sent_df = within(sent_df,
emotion<- factor(emotion, levels=names(sort(table(emotion),
decreasing=TRUE))))
```

A histogram plot of the distribution of emotions is helpful to get a perceptual idea of sentiment distribution within the Twitter corpus. Figure 2 gives a demonstration of this distribution plot, using the `ggplot()` function.

```
# plot distribution of emotions

ggplot(sent_df, aes(x=emotion)) +
geom_bar(aes(y=..count.., fill=emotion)) +
scale_fill_brewer(palette="Dark2") +
xlab("emotion categories") + ylab("number of tweets") +
ggtitle("Sentiment Analysis of tweets about\n Women's
Reservation \n(classification by emotion)") + theme (text =
element_text ( size = 12 , family="Impact"))
```

Similar to sentiment distribution, we may have a polarity distribution of the corpus from the histogram plotting of the polarity values of the data frame `sent_df` (Figure 3).

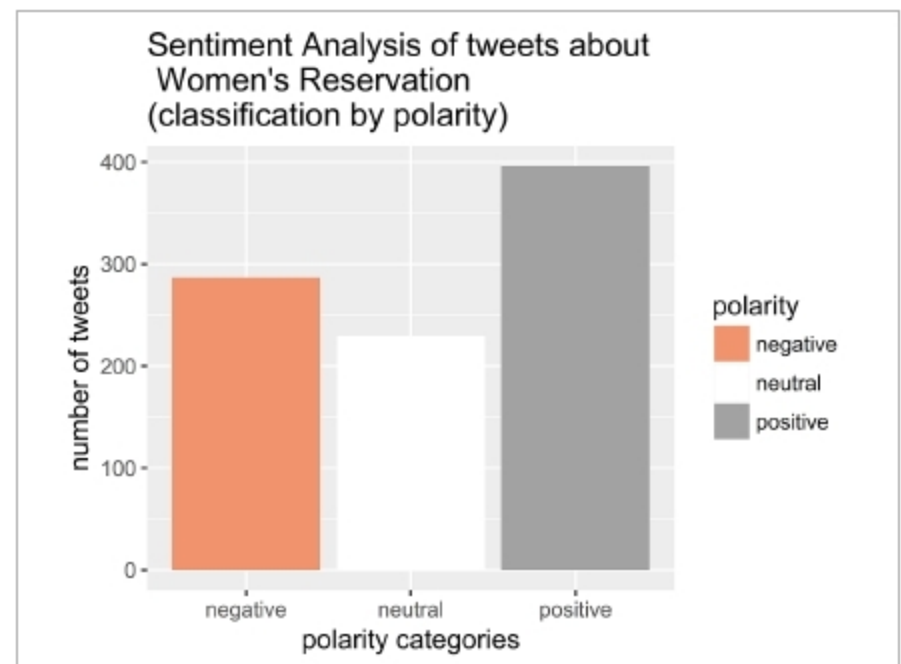


Figure 3: Histogram plot of distribution of polarity

```
ggplot(sent_df, aes(x=polarity)) +
geom_bar(aes(y=..count.., fill=polarity)) +
scale_fill_brewer(palette="RdGy") +
xlab("polarity categories") + ylab("number of tweets") +
ggtitle("Sentiment Analysis of tweets about \n Women's
Reservation \n(classification by polarity)") + theme ( text =
element_text ( size = 12 , family = "Impact" ) )
```

It can be seen that both these frequency distribution histograms are quite useful graphical depictions of the sentiments of 1500 tweets spread over the Indian subcontinent. As expected, the largest number can be categorised as unclassified, but the remaining values are quite helpful to get a clear idea of an individual's feelings about the subject of debate. The analysis is based on some predefined emotion categories and the vocabulary of the test database is also limited; so there are ample opportunities for improvement. **END** 🐧

References

- [1] <https://social.msdn.microsoft.com/Forums/sqlserver/en-US/9f87b253-5f0d-454c-a319-f5ea4b2ee151/10-class-emotion-classification-in-text?forum=MachineLearning>
- [2] https://github.com/ManoharSwamynathan/Sentiment_Analysis/tree/FirstCommit/data
- [3] <https://manoharswamynathan.wordpress.com/2016/08/06/sentiment-analysis/>
- [4] <https://rhandbook.wordpress.com/tag/sentiment-analysis-using-sentiment-package-2/>

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